

CCM Research Idea Sketch

First paper: <https://arxiv.org/pdf/2212.03827> (1)

Methodology:

- Given questions $q_1, q_2, \dots, q_n$ where each q_i could be something like "333+25 = 358?"
- Generate statements $x_1^+, \dots, x_n^+$ where each $x_i = q_i + \text{"Yes"}$ and $x_1^-, \dots, x_n^-$ where $x_i = q_i + \text{"No"}$
- Feed the statements to a LLM and find the activations $\phi(x_1^+), \phi(x_1^-), \dots, \phi(x_n^+), \phi(x_n^-)$
- Modify the results by having $\bar{\phi}(x_i^+) = \frac{\phi(x_i^+) - \mu^+}{\sigma^+}$, "" so that the suffix gets taken out
- Now start training a linear probe $p_{\theta, b}(\bar{\phi}) = \sigma(\theta^T \bar{\phi} + b)$ giving probability something is true
- Look at $(\bar{\phi}(x_i^+) + \bar{\phi}(x_i^-) - 1)^2$ which is consistency (false + truth - 1 should equal 0)
- Look at $\min(\bar{\phi}(x_i^+), \bar{\phi}(x_i^-))$ for confidence
- Minimize consistency + confidence
 - Technically we don't know if $p = 1$ is true or false but we can identify this with conjunctions

Result: **Contributes to the ELK problem by helping identify when the model output is true or false**

Advantages:

- Useful for domains where the model knows the ground truth but humans don't
- Identifies deceptive models that say what they don't believe from honest models (regardless if the honest models arrive at the correct answer)

Present issues:

- CCM is sensitive to prefixes like "yes" or "no" / "right" or "wrong" / stuff like "not"
 - <https://arxiv.org/pdf/2511.02089> (2)
 - <https://leonardbereska.github.io/blog/2024/mechinterpreview/> (3)
- <https://arxiv.org/pdf/2501.16496#page=20&zoom=100,62,304> said this was a big issue as well
 - Datasets made by hand too slow
 - Often only has right/wrong